

Compute estimate — training a Qwen3-class kinetic model to beat baseline from scratch

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Status: decision support, not an ADR — no decision has been made to pursue this path.

Scope: what it would cost, in dollars and wall-clock, to train a kinetic-architecture base model large and heavily-trained enough to independently beat Qwen3-class baselines on public benchmarks — via full from-scratch pretraining, or via teacher distillation — priced across RunPod GPU options.

Author context: produced by an agent session in worktree `kinetic-ai-runpod-estimate-76abc3` on operator request. Not reviewed or acted on.

1 This contradicts two things already on record

`docs/PRD.md`’s stated success criterion is that the system **beats Qwen3-1.7B** (the active-parameter-matched baseline) and is credited against Qwen3-4B — but the mechanism is the **equilibrium council aggregating existing smaller models at inference time**, not out-training Qwen3 via a bigger pretraining run. The same document’s Non-goals section states plainly:

Training a competitive base model from scratch, which the available compute rules out.

That conclusion is not a hand-wave. It is the direct result of two experiments this project already ran and closed, recorded in `research/memory/findings.md`:

- **F44 / F45** — kinetic tying is a parameter $\leftrightarrow$ compute *trade*, never a compute *saving*. At matched compute, tying yields 2.70x fewer resident parameters for $\sim 96\%$ of an explicit model’s quality. To reach full parity with an explicit model at *matched parameters*, tying costs **4.92x more compute** (F44). There is no regime in which tying reduces the FLOPs needed to reach a target quality bar.
- **F51** — converting a pretrained (already-trained) model into tied form fails at any gentleness tested; no cheap conversion shortcut exists.
- **F53** — the knowledge-distillation pilot (student trained from scratch against Qwen2.5-1.5B-Instruct as teacher) made results **worse**, not better: cross-entropy alone reached 482.96 held-out perplexity; adding temperature-softened logit distillation reached 493.39 — a -2.2% change, the wrong direction. This closed “cheap distillation” as a shortcut to scale, with two open caveats noted below (§4).

This document quantifies exactly what “requires a pretraining budget” costs, so the scale of the non-goal is visible in dollars rather than asserted.

2 Method

Standard compute-scaling estimate: $C \approx 6ND$ FLOPs, where N = parameters and D = training tokens (the usual forward + backward FLOPs approximation used in scaling-law literature). Achieved GPU throughput is assumed at **35% model-FLOPs-utilization (MFU)** of each GPU’s public dense-BF16 TFLOPS spec. This is an assumption, not a measurement — this project’s own discipline (SPEC 0022’s preflight GO rule) is to verify throughput before trusting it with real budget, and that discipline should apply here before any of these numbers are spent against.

Kinetic tying gets the **4.92x compute tax** (F44) applied whenever the target is a resident parameter count matched to a Qwen3 size — because tying does not save compute, it only trades compute for a parameter saving.

Two token budgets are modeled per target size:

- **Chinchilla floor** (20 tokens/param) — cheap, but not remotely competitive with a heavily-trained modern model; this is roughly the regime SPEC 0022’s current ~10B-token programme already sits in.
- **Competitive / overtrained** (~1,500 tokens/param) — anchored to real modern small-model practice (Llama-3-8B: 15T tokens / 8B params $\approx$ 1,875 tokens/param). This is the realistic bar to actually contend with a model like Qwen3, which was itself heavily overtrained relative to Chinchilla-optimal.

3 Headline numbers — Qwen3-1.7B-class, kinetic-tied, competitive budget

Scenario	Tokens	FLOPs
Chinchilla floor, explicit-equivalent	34B	3.47e20
Chinchilla floor, kinetic-tied ($\times 4.92$ tax)	34B	1.71e21
Competitive, explicit-equivalent	2.55T	2.60e22
Competitive, kinetic-tied ($\times 4.92$ tax)	2.55T	1.28e23

Cost and time by RunPod GPU — competitive budget, kinetic-tied (the actual ask)

GPU	Achieved TFLOPS (35% MFU)	GPU-hours	Community \$	Secure \$
RTX 5090	73	486,000	\$335,400	\$481,200
A100 SXM 80GB	109	325,600	\$452,600	\$517,700
H100 SXM	346	102,700	\$276,300	\$337,900
H200 SXM	346 (assumed equal to H100)	102,700	\$368,700	\$471,400
B200 (Secure only)	787	45,150	—	\$306,500

H100 is the cheapest single-GPU-equivalent option at $\approx$ \$276K. Even the fastest option (B200) needs ~45,150 GPU-hours; no RunPod listing offers more than 8 GPUs per pod today, so reaching this in weeks rather than months would require a genuinely large multi-node cluster — well beyond anything discussed in the RunPod migration estimate for the current 1B-scale run.

For calibration, the **Chinchilla-floor** / **kinetic-tied** case (not competitive — a sanity check only) costs roughly \$4–6 and finishes in under 90 minutes on H100, consistent with where SPEC 0022 already sits. The gap between “cheap” and “actually competitive” is about five orders of magnitude — the concrete shape of the PRD’s “available compute rules out” line.

Explicit-equivalent floor (no kinetic tax) — for calibration only

Divide the table above by 4.92. This does not deliver the parameter-count saving kinetic tying is for, it only establishes the compute floor for an explicit model at the same budget.

GPU	Community \$	Secure \$
RTX 5090	\$68,171	\$97,810
A100 SXM 80GB	\$92,000	\$105,226
H100 SXM	\$56,150	\$68,675
B200	—	\$62,300

Scaling to Qwen3-4B

Roughly 5.54x the 1.7B FLOPs ($N \cdot D$ scaling at the same token/param ratio). Competitive, kinetic-tied cost lands around **\$1.5M–\$2.7M** across the same GPU spread.

4 The “via teacher” path

F53 already tested this and found it made things worse, at the token regime tried (16.4M tokens, an extreme early-stage regime). Two honest caveats the project itself flagged, neither yet retested:

- Only one KD configuration was tried (distillation weight 1.0, temperature 2.0); a sweep might behave differently.
- 16.4M tokens may be too early for a from-scratch student to use a 151k-way soft teacher distribution at all — published KD successes distill into pruned-from-teacher initializations that start near the teacher, not into random weights.

If pursued regardless, distillation currently has **no proven compute discount** to offset against, and it adds real cost: running the teacher’s forward pass alongside the student’s training adds roughly $2N_{\text{teacher}}D$ FLOPs on top of the student’s own $6N_{\text{student}}D$. For a 7B teacher against a 1.7B student, that is $\approx +137\%$ **extra compute** — making the teacher path more expensive than from-scratch under current evidence, not less.

Recommendation if this path is still of interest: run a proper (not “cheap”) KD sweep first — a few configurations, a larger token count than F53’s 16.4M, at roughly the same ~ 1 -day 5090 cost as the original pilot — before committing any of the six-figure budgets above to either path. That is a near-\$0 experiment (already-owned hardware) that could overturn F53, versus a \$270K+ commitment that currently assumes it already has.

5 What this means for the programme

Under current evidence, neither shortcut (conversion, cheap distillation) survives, and the full-budget path costs 5+ orders of magnitude more than the current SPEC 0022 programme. The

PRD’s actual success criterion — beating Qwen3-1.7B via the equilibrium council rather than via a bigger base model — remains the validated, affordable path. This document exists so that choice is visible as a quantified tradeoff rather than an assumption, and so a future decision to revisit the non-goal (if RunPod economics or a reversed KD finding change the calculus) has real numbers to start from.

6 Before spending against any number here

1. These are extrapolated estimates (6ND scaling law, 35% MFU assumption), not measurements. Re-verify with a short preflight on the target GPU before committing real budget, per this project’s own preflight discipline.
2. Confirm RunPod pricing/availability at execution time — queried live 2026-08-30 for the companion `docs/runpod-migration-estimate.md`; prices move.
3. Run the proper KD sweep (§4) before assuming the teacher path is closed for good — F53 tested one configuration at one very early token count, not the design space.
4. This document evaluates a strategy the PRD lists as a non-goal. Any decision to pursue it should be an explicit operator redirection of PRD scope, not an implicit one.